Routledge
Taylor & Francis Group

Innovations in Education and Teaching International

ISSN: 1470-3297 (Print) 1470-3300 (Online) Journal homepage: www.tandfonline.com/journals/riie20

University students' perceptions of generative AI for critical thinking and creativity: The influence of self-efficacy and disciplinary differences

Atiyah A. Alghamdi

To cite this article: Atiyah A. Alghamdi (06 Dec 2025): University students' perceptions of generative AI for critical thinking and creativity: The influence of self-efficacy and disciplinary differences, Innovations in Education and Teaching International, DOI: 10.1080/14703297.2025.2600476

To link to this article: https://doi.org/10.1080/14703297.2025.2600476

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Published online: 06 Dec 2025.

Submit your article to this journal

Article views: 1903

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=riie20

INNOVATIONS IN EDUCATION AND TEACHING INTERNATIONAL
https://doi.org/10.1080/14703297.2025.2600476

OPEN ACCESS Check for updates

University students' perceptions of generative AI for critical thinking and creativity: The influence of self-efficacy and disciplinary differences

Atiyah A. Alghamdi

Self-Development Skills Department, King Saud University, Riyadh, Kingdom of Saudi Arabia

ABSTRACT

The rapid integration of generative artificial intelligence tools, such as ChatGPT, into higher education has prompted debates about their influence on students' cognitive and creative capacities. This quantitative, cross-sectional study surveyed 405 undergraduates to explore how academic discipline (STEM vs. non-STEM) and AI-related self-efficacy shape these perceptions. Results indicate that students hold moderately positive views of Generative AI's impact on critical thinking and creativity. Independent-samples t-tests showed that STEM students rated AI's support for critical thinking significantly higher than non-STEM peers, whereas disciplinary differences in creativity perceptions were non-significant. Simple linear regression revealed that self-efficacy in AI use is a significant predictor of perceived critical-thinking enhancement and creativity support. These findings underscore the central role of students' confidence in leveraging AI tools as an educational asset. We conclude that developing targeted interventions to bolster AI self-efficacy will maximize AI's potential to foster higher-order thinking and innovative idea generation.

KEYWORDS

ChatGPT; critical thinking; creativity; self-efficacy; generative AI; STEM

Introduction

Generative Artificial Intelligence (GenAI) refers to machine learning models designed to create new content, including text, images, audio, and even code, by learning patterns from existing datasets (Chan & Hu, 2023). These AI models rely on deep learning techniques, such as Generative Adversarial Networks (GANs) and Transformer-based architectures, to produce highly coherent and contextually relevant outputs (Yusuf et al., 2024).

Higher educational institutes have adopted GenAI for varied applications, such as personalised tutoring, automated grading, academic writing, and research support, among others (Marengo et al., 2024). Evidence suggests that the use of AI tools can lead to higher cognitive learning behaviour since it leads learners to engage in higher-order thinking skills, such as evaluation and synthesis, promoting more than mere recall (Gonsalves, 2024). In STEM education, AI-powered

CONTACT Atiyah A. Alghamdi aalatia@ksu.edu.sa Self-Development Skills Department, King Saud University, Riyadh 11451, Kingdom of Saudi Arabia

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

applications help students diagnose complex issues by providing explanations and alternate solutions step by step (Shanto et al., 2024). Despite the potential for AI as an educational tool, its use raises ethical issues concerning its usage, including plagiarism, misinformation, and students depending on AI to produce their own work (Yusuf et al., 2024).

Researchers report that GenAI tools can enhance critical thinking skills, specifically in structured argumentation, by offering real-time feedback and prompting students to articulate claims, supports, and counterarguments more thoroughly (Noroozi, Alqassab, et al., 2025). However, the AI-generated texts tended to be more similar to one another, raising concerns about creative homogeneity. This echoes broader critiques that while AI may boost generative ideation, it may also limit true novelty unless mediated by human critical reflection (Wang et al., 2025).

Banihashem et al. (2024) reported that while AI excels at providing structural and descriptive feedback, human peers offer more profound insights into problem identification and reasoning, underscoring the need for hybrid feedback models that combine artificial and human cognition. They emphasise the importance of pedagogical frameworks that guide AI integration in ways that preserve academic integrity and foster higher-order thinking. The work of Banihashem et al. (2025) proposes a model of Hybrid Intelligent Feedback, where AI-generated suggestions are embedded within dialogic, reflective learning environments to promote argumentation-based reasoning and self-regulated learning.

This aligns with broader concerns raised by Noroozi, Alqassab, et al. (2025), who cautioned against technological determinism and advocate for human-AI collaboration grounded in educational theory. These scholars highlight that the impact of GenAI on creativity is context-dependent: while it may stimulate divergent thinking in ideation phases, it risks convergent homogenisation if students lack metacognitive strategies to critically refine AI outputs (Kirkpatrick, 2023). Taken together, these studies suggest that the educational value of GenAI lies not in its standalone capabilities but in how it is pedagogically orchestrated to support argumentation, critical inquiry, and creative synthesis across disciplinary contexts.

Although a growing body of research has examined the pedagogical implications of GenAI, there is still a research gap. For example, most of the studies have been either conceptual or exploratory, focusing primarily on ethical concerns, instructional potential, or educators' perspectives (Holmes et al., 2022; Walter, 2024; Yusuf et al., 2024). Empirical work that has quantitatively investigated how students themselves perceive AI's influence on higher-order cognitive processes remains limited. Few studies have explored how such perceptions vary across disciplinary contexts or how students' self-efficacy in AI use mediates these views. In addition, previous research has often treated critical thinking and creativity as separate constructs or investigated them indirectly through academic performance indicators (Premkumar et al., 2024). Banihashem et al. (2024, 2025) highlighted the importance of argumentation and feedback in AI-supported learning, yet there is still a lack of understanding about how learners' confidence in using AI interacts with their disciplinary orientations to shape these cognitive and creative perceptions. The current study addresses this research gap in several ways. First, it empirically investigates the intersection of generative AI, critical thinking, and creativity from the learner's perspective, using a large quantitative dataset. Second, it introduces self-efficacy in AI use as a key psychological factor that predicts students' perceptions of AI's cognitive and creative affordances. Finally,

the study captures how disciplinary epistemologies influence students' interpretations of AI's educational value by comparing between STEM and non-STEM students.

Theoretical framework

Critical thinking in educational contexts extends beyond generic problem-solving to include the capacity for argumentation-based reasoning, evaluation of evidence, and meta-cognitive regulation of thought processes (Kuhn, 1991). In AI-mediated learning environments, these dimensions may be reshaped as students interact with large-scale language models that offer pre-structured arguments or suggest solutions. Such technologies can act as cognitive scaffolds (Granda et al., 2024). Yet, there is a risk that they may also foster cognitive passivity if students rely on AI-generated content without engaging in reflective critique (Holmes et al., 2022).

Creativity, by contrast, is typically conceptualised through the lens of divergent and convergent thinking (Guilford, 1967). Divergent thinking involves generating multiple, novel, and original ideas, whereas convergent thinking emphasises narrowing down ideas to find optimal solutions. GenAI tools, such as ChatGPT, have been shown to stimulate idea generation (divergent thinking). However, their algorithmic nature often gravitates towards patterned, norm-based outputs, which may constrain truly novel ideation unless mediated by user criticality and metacognitive awareness (Kirkpatrick, 2023). This duality necessitates careful pedagogical framing to ensure students actively shape, rather than passively accept, AI-generated suggestions.

Furthermore, disciplinary epistemologies deeply influence how critical and creative thinking are both defined and operationalised. In STEM disciplines, creativity may be viewed through the lens of technical innovation, system optimisation, or procedural efficiency – often aligning with convergent thinking processes. In contrast, humanities and social sciences tend to value interpretive creativity, ambiguity tolerance, and the reconfiguration of meaning – hallmarks of divergent thinking (Karwowski et al., 2020). This epistemological divergence impacts how students in different academic domains interpret the utility and limitations of GenAI tools.

The present study is anchored in Social Cognitive Theory (SCT) (Bandura, 1986) and Constructivist Learning Theory (Jonassen, 1999), which together provided the conceptual foundation for both the research design and the interpretation of findings. SCT emphasises the reciprocal interactions between personal beliefs, environmental influences, and behaviours. Within this study, the construct of self-efficacy was critical, as it represents students' belief in their ability to effectively use generative AI tools. This perspective informed the inclusion of a validated self-efficacy scale, ensuring that the survey not only captured students' perceptions of AI but also their confidence in engaging with such technologies. The methodological decision to conduct regression analyses examining self-efficacy as a predictor of perceptions was a direct application of SCT, which posits that self-beliefs significantly shape cognitive engagement and learning outcomes.

Constructivist Learning Theory provided the second layer of theoretical scaffolding, emphasising that learning is an active process of meaning-making shaped by prior knowledge and context (Jonassen, 1999). In this study, disciplinary orientation (STEM vs. non-STEM) was conceptualised as a contextual factor that influences how students interact with generative AI tools. Constructivism guided the comparative methodological design, particularly the

decision to analyse disciplinary differences. By contrasting how students from different knowledge traditions perceive AI's role in critical and creative cognition, the study aligns with constructivist assumptions that epistemological orientations shape cognitive engagement and learning practices.

Research questions

This study will focus on the following research questions:

(1) How do higher education students perceive the impact of generative AI on their critical thinking skills?
(2) How do students perceive the role of generative AI in enhancing or hindering creativity?
(3) How do students' perceptions of generative AI as a tool for critical thinking and creativity vary across different academic disciplines?
(4) What is the relationship between students' self-efficacy in AI use and their perceptions of generative AI's role in critical thinking and creativity?

Methodology

Research design

This study adopts a quantitative survey-based research design to investigate higher education students' perceptions of generative AI as a tool for critical thinking and creativity. A cross-sectional survey design was employed to collect numerical data that enables statistical analysis of relationships among key variables (Soomro et al., 2024). This study underpinned Constructivist Learning Theory and Social Cognitive Theory (Bandura, 1986), particularly the construct of self-efficacy, (i.e. students' belief in their capacity to effectively use generative AI tools). This theoretical lens informed the inclusion of validated scales for AI-related self-efficacy, recognising that behaviour is shaped by reciprocal interactions between personal confidence, digital competence, and the academic environment.

Population and sampling

The target population consists of first-year university students enrolled in various academic disciplines. Academic disciplines were dichotomised into STEM (Science, Technology, Engineering, and Mathematics) and non-STEM (including Social Sciences, Humanities, and Business-related disciplines) categories. This dichotomy is supported by prior educational research that demonstrates how disciplinary epistemologies shape students' engagement with technology and cognitive tasks (Becher & Trowler, 2001).

The required sample size was estimated using G*Power 3.1 software for multiple regression analysis. Based on Cohen's (2013) guidelines, assuming a medium effect size ($f^2 = 0.15$), an alpha level (α) of 0.05, and a power ($1 - \beta$) of 0.80, a minimum of 250–300 participants is recommended to ensure reliable statistical analysis. Two hundred and forty-five male (60.5%) and 160 female (39.5%) participants completed the survey. Although the

study focused on first-year undergraduates, participants' ages ranged from 17 to 24 years. This broader distribution reflects the unique features of higher education in the Saudi context. Students often enter university at different stages due to several factors, including preparatory (foundation) year programmes that are required in many institutions, delayed enrolment following high school graduation, and varying entry pathways such as transfer from diploma programmes or re-entry after a gap period. Most of the participants were 18 years ($n = 177$, 43.7%) and 19 years old ($n = 159$, 39.3%), followed by 20 years ($n = 45$, 11.1%). Four participants (1.0%) were 17 years old, four (1.0%) were 23, and one (0.2%) was 24. Whereas two of the participants did not mention their age. The participants belonged to health, medical, science, business, and humanities-related academic domains. About two-third ($n = 298$, 73.6%) of them belonged to STEM-related disciplines, while one-third ($n = 107$, 26.4%) of them belonged to non-STEM domains. In the Saudi higher education context, STEM enrolments are disproportionately higher due to national priorities in science and technology and state investment in health and engineering programmes compared to the humanities and social sciences. The sample in this study reflects this institutional reality, with 73.6% of participants coming from STEM disciplines. Table 1 summarises respondents' demographic characteristics.

Measures

The survey instrument consisted of some demographic items and validated scales adapted from prior research to measure key variables of the study. A 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree) was used for all construct-related questions. Table 2 highlights the three scales related to the main variables of this study. The survey items were reviewed by three domain experts in education and technology to ensure alignment with the research objectives.

Study procedures

Data were collected via an online survey administered through Google Forms. The survey link was distributed through institutional emails, student discussion forums, and social

Table 1. Respondents' demographic characteristics.

Group/Characteristic	*n* (%)
Gender	
Male	245 (60.5%)
Female	160 (39.5%)
Age	
17 years	4 (1.0%)
18 years	177 (43.7%)
19 years	159 (39.3%)
20 years	45 (11.1%)
21 years	8 (2.0%)
22 years	5 (1.2%)
23 years	4 (1.0%)
24 years	1 (0.2%)
Discipline	
STEM	298 (73.6%)
Non-STEM	107 (26.4%)

Table 2. Survey sections and measurement scales.

Construct	Number of Items	Source
Self-Efficacy in AI Use	5	Adopted from Falebita and Kok (2024).
Perceived Impact of AI on Critical Thinking	6	Inspired by The California Critical Thinking Disposition Inventory (CCTDI) of Facione et al. (1994)
Perceived Impact of AI on Creativity	7	Inspired by Runco's Creativity Assessment Battery (2012)

media groups. Participation in the study was voluntary, anonymous, and confidential, with an informed consent statement provided to study participants at the beginning of the survey.

Ethical considerations

The study adheres to ethical research guidelines, including obtaining approval from an institutional review board (IRB) at King Saud University before data collection. Participants were informed about the study's purpose, voluntary participation, and confidentiality. No personally identifiable information was collected to ensure participants' privacy. Moreover, the data will be reported in aggregated form only.

Results

This section reports the results of the current study in response to the above-stated four research questions.

Perceived impact of generative AI on students' critical thinking

Six items were formatted on a five-point Likert scale to capture various dimensions of participants' critical thinking, including analytical reasoning, synthesis, questioning of assumptions, and reliance on personal versus AI-generated cognition. Table 3 shows item-wise descriptive statistics for this construct.

The descriptive statistics showed that the mean scores for the items ranged from 3.55 to 3.72. These values indicate a moderately positive perception of AI's role in enhancing critical thinking. The overall mean score for the construct was 3.63 (SD = 0.64), indicating that, on average, students somewhat agree that GenAI tools, such as ChatGPT, positively

Table 3. Item-wise descriptive statistics of perceived impact of AI on critical thinking.

S#	Statement of items within the Scale	Mean	Std. Deviation
1	Generative AI helps me analyse information more critically.	3.67	0.949
2	Using AI tools enhances my ability to evaluate different perspectives.	3.63	0.991
3	AI-generated content encourages me to question my understanding of a topic.	3.58	1.006
4	Generative AI improves my ability to synthesise multiple sources of information.	3.72	0860
5	AI usage makes me more reliant on external sources rather than my own critical thinking. (Reverse-coded item)	3.55	0.827
6	AI-assisted learning encourages me to ask deeper follow-up questions about complex topics.	3.62	0.966

Note: $N = 405$.

influence their critical thinking abilities. The findings reflect a moderately favourable perception of GenAI's impact on critical thinking. Students seemed to recognise AI's utility in analysing and evaluating information, synthesising diverse sources, and encouraging reflective questioning.

Perceived impact of generative AI on students' creativity

Seven items were formatted on a five-point Likert scale to assess various facets of creativity, including idea generation, flexibility in thinking, expression of novel ideas, and the perceived limitations imposed by AI use. Table 4 shows item-wise descriptive statistics for this construct.

As can be seen from Table 4, the mean values for individual items ranged from 3.14 to 3.39. These values reflect general agreement, though not strongly so. The results suggest that students moderately agreed that GenAI tools, such as ChatGPT and similar platforms, support their creative thinking processes. Specifically, students acknowledged that AI aids in idea generation, enhances problem-solving creativity, and assists in expressing ideas in diverse ways. Overall, the average of the score was 3.3192 with a standard deviation of .850. This indicates that, on average, Saudi undergraduate students perceived that the use of GenAI, such as ChatGPT, is positive to enhance their creativity.

Disciplinary differences in students' perceptions

An independent t-tests was conducted with *academic discipline (STEM/Non-STEM)* as independent variable (IV) and *critical thinking and creativity* as dependent variables (DV) to investigate discipline-based differences (STEM and non-STEM) and determine whether there is statistical evidence that the associated population means are significantly different (Field, 2018). The results of the t-test revealed that there was a significant difference in the critical thinking perceptions between the students from STEM-related disciplines ($M = 3.701$, $SD = .588$) and non-STEM disciplines ($M = 3.427$, $SD = .734$); $t(403) = 3.858$, $p < .001$. The results suggested that the university students enrolled in STEM-related fields were more positive regarding the role of GenAI tools in enhancing students' critical thinking than their counterparts studying in non-STEM related fields. This implies that STEM students perceive GenAI tools more beneficial in fostering analytical reasoning, evaluation of perspectives, and synthesis of information. This may reflect the alignment of AI tools with computational, logic-based tasks prevalent in STEM education.

Table 4. Item-wise descriptive statistics of perceived impact of AI on students' creativity.

S#	Statement of Items within the Scale	Mean	Std. Deviation
1	Generative AI helps me generate original and creative ideas.	3.32	1.202
2	Using AI in academic tasks enhances my ability to think outside the box.	3.33	1.189
3	AI-generated suggestions stimulate my creative problem-solving skills.	3.32	1.181
4	Relying on AI-generated content limits my personal creativity. (Reverse-coded item)	3.14	1.117
5	I use AI to explore new ways of expressing my ideas.	3.39	1.196
6	AI tools help me combine different concepts in novel ways to create unique ideas.	3.36	1.176
7	Using AI enhances my ability to generate multiple solutions to a problem.	3.38	1.216

Note: $N = 405$.

Table 5. Independent sample t-test results.

		t	df	Sig.	Mean Difference	Std. Error Difference
Critical_Thinking	Equal variances assumed	3.858	403	.000	.27399	.07102
	Equal variances not assumed	3.481	157.675	.001	.27399	.07871
Creativity	Equal variances assumed	1.062	403	.289	.10179	.09582
	Equal variances not assumed	1.072	190.377	.285	.10179	.09496

On the other hand, the results revealed that there was no significant difference between the digital competence of male ($M = 4.124$, $SD = .671$) and female students ($M = 4.178$, $SD = .572$); $t(403) = -1.062$, $p = .289$ (see Table 5 for details). It indicated that STEM-students and non-STEM students held almost the same perceptions regarding the role of AI tools on creativity. This finding suggests a convergence of attitudes across disciplines, possibly due to the increasingly interdisciplinary nature of creativity and the broadening applicability of AI tools in idea generation and problem-solving across academic domains.

Influence of self-efficacy in AI use on students' perceptions

Five items were formatted on a five-point Likert scale to assess university students' self-efficacy of using artificial intelligence. These items along with their descriptive statistics are listed in Table 6.

The mean values for individual items ranged from 3.37 to 3.44. Overall, the average score for self-efficacy was 3.398, with standard deviation of 1.032. This implies that, on average, Saudi undergraduate students feel moderately competent in using generative AI tools. A simple linear regression analysis was performed for each of the two dependent variables (critical thinking and creativity). A scatterplot showed that the relationship between AI self-efficacy and critical thinking was positive and linear and did not reveal any bivariate outliers. The homoscedasticity and linearity were established. There was no need to check the collinearity because there was only one predictor variable. The correlation between AI's self-efficacy and critical thinking was statistically significant, $r(404) = .604$, $p < .001$.

The regression equation for predicting the perceived AI's role in critical thinking from the level of AI self-efficacy was $\hat{y} = 2.355 + 0.375\ (X)$. Table 7 displays the result of the regression analysis. The r^2 for this equation was .364; that is, 36.4% of the variance in critical thinking was predictable from the level of AI self-efficacy. This is a medium (moderate) relationship (Cohen, 2013). The bootstrapped 95% confidence interval* for the slope to predict perceived critical thinking from self-efficacy ranges from .326 to .423; thus, for each one unit of increase in AI self-efficacy, critical thinking increases about .33 to

Table 6. Item wise descriptive statistics of AI self-efficacy.

S#	Statement of Items within the Scale	Mean	Std. Deviation
1	I feel confident in my ability to use AI tools effectively to accomplish my tasks.	3.42	1.167
2	I believe I have the necessary skills and knowledge to get the most out of AI tools.	3.39	1.156
3	I feel in control when working with AI tools and can achieve the	3.37	1.153
4	outcomes I desire.	3.38	1.145
5	I am comfortable experimenting and troubleshooting when using AI tools.	3.44	1.173

Note: $N = 405$.

Table 7. Simple linear regression results predicting critical thinking with self-efficacy ($n = 405$).

Variables	*B*	*SE*	*β*	*t*
(Constant)	2.355	.088		26.892*
Self-Efficacy	.375	.025	.604	15.195*
R		.604		
R^2		.364		
Adjusted R^2		.363		
F		230.893*		

*$p < .001$.

.42 points. These results indicate that students whose self-efficacy of AI usage is high are more likely to perceive AI tools as enablers of critical thinking.

Similarly, a scatterplot showed that the relationship between AI self-efficacy and creativity was positive and linear and did not reveal any bivariate outliers. The homoscadicity and linearity were established for the model. There was no need to check the collinearity in this case because there was only one predictor variable (creativity). The correlation between AI's self-efficacy and creativity was statistically significant, $r(405) = .807$, $p < .001$.

The regression equation for predicting the perceived AI's role in creativity from the level of AI self-efficacy was $\hat{y} = 1.061 + 0.665$ (X); see Table 8 for the detailed results of regression analysis. The r^2 for this equation was .651; that is, 65.1% of the variance in creativity was predictable from the level of AI self-efficacy. This is a large (strong) relationship (Cohen, 2013). The bootstrapped 95% confidence interval for the slope to predict perceived creativity from self-efficacy ranges from .617 to .712; thus, for each one unit of increase in AI self-efficacy, creativity increases about .62 to .71 points. These results suggest that students who feel more competent in using AI tools are more likely to perceive them as enablers of creative ideation.

Discussion

The results of the study revealed that undergraduate students perceive GenAI tools, such as ChatGPT, as a support in developing their critical thinking skills. They primarily view AI as beneficial in enhancing analysis, synthesis, and evaluative thinking. At the same time, there remains a subtle awareness of its possible limitations, particularly in fostering dependency. These findings align with Granda et al. (2024) and Gonsalves (2024). Their work suggests that AI tools can function as cognitive scaffolds – prompting students to reconsider perspectives, synthesise information, and formulate more profound questions.

Table 8. Simple linear regression results predicting creativity with self-efficacy ($n = 405$).

Variables	*B*	*SE*	*β*	*t*
(Constant)	1.061	.086		12.318*
Self-Efficacy	.665	.024	.807	27.408*
R		.807		
R^2		.651		
Adjusted R^2		.650		
F		751.199*		

*$p < .001$.

These findings also substantiate theories in the literature that argue AI can serve as both a facilitator and a potential inhibitor of higher-order thinking (Chan & Hu, 2023; Premkumar et al., 2024).

On the other hand, Gerlich (2025) highlights the phenomenon of cognitive offloading, where frequent AI use may lead to a decrease in users' engagement in deep reasoning tasks. This aligns with concerns raised in the present study about over-reliance on AI-generated suggestions, particularly among students with lower self-efficacy. To mitigate this, educational strategies must promote critical digital literacy. Holmes et al. (2022) also noted the dual role of AI. They suggested that the role of AI in education be continuously evaluated to determine whether it leads to cognitive passivity. The moderate level of agreement among students indicate that while they view AI as beneficial, they do not believe that it is transformational in its current use. This highlights the need for pedagogical frameworks designed around AI to promote active and reflective engagement rather than passive consumption.

The study participants also acknowledged AI's potential in generating novel ideas, but again with moderate interest. The results suggest that students moderately agree with the support of GenAI tools in their creative thinking processes. Specifically, they acknowledge that AI aids in idea generation, enhances problem-solving creativity, and facilitates the expression of ideas in diverse ways. However, the relatively modest mean scores (mostly close to 3.30) suggest a somewhat cautious optimism. This middle-ground stance might stem from the dual nature of AI's functionality – both as an enabler and a constraint (Premkumar et al., 2024). It is consistent with recent evidence by Kirkpatrick (2023) who found that while AI can enhance ideation, it may also boost convergence towards formulaic outputs. While GenAI can simulate divergent thinking, it lacks the embodied cognition and emotional depth that characterise authentic human creativity. This philosophical perspective reinforces the study's finding that students perceive AI as a tool for idea generation but remain cautious about its capacity for originality (Lockhart, 2025).

A significant finding of this study was the large difference in critical thinking perceptions of STEM and non-STEM students. More STEM students believe AI to be a tool that supports analytical reasoning and problem-solving. This aligns with broader trends in the literature, which show that the STEM disciplines – typically oriented towards structured problem-solving – more easily take up AI applications within coursework and assessments (Xu & Ouyang, 2022). Previous literature also showed that technological competence and practices are often contextually influenced by academic disciplines (Soomro et al., 2015).

On the other side, creativity perceptions did not vary at a significant level between STEM and non-STEM students. This finding is indeed intriguing because it implies that there is some convergence among students regarding the creative potential of AI across disciplines. A possible explanation for this can be grounded in Karwowski et al'.s (2020) assertion that creative tasks have become increasingly cross-disciplinary within higher education; design thinking and innovation are highlighted in both curricula, technical and non-technical. Also, across disciplines, the manner in which students explore AI may be similar due to generative tools' ubiquity and availability of such platforms, such as ChatGPT and Canva.

Learner background characteristics clearly shape how students engage with GenAI. The study involved first-year undergraduates who may vary considerably in prior exposure to AI tools, confidence, and discipline-specific skills. Such factors likely

influenced their interaction with GenAI. Previous research on AI-augmented learning underscores this. For example, Sichterman et al. (2025) observe that studies of AI-supported peer learning routinely account for students' prior knowledge, skill levels, and experience with peer feedback. These academic characteristics help determine baseline competencies. Our findings mirror and extend recent empirical results on GenAI and peer feedback. Noroozi, Haddadian, et al. (2025) examined graduate students who had instruction in ethical GenAI use before a peer-review task. They report that just over half of students chose not to use GenAI for peer feedback, primarily because they believed they would learn more by completing the task independently. Those who did engage GenAI found it moderately helpful: notably, GenAI users provided significantly more high-level feedback suggestions (e.g. on structure and arguments) than non-users. This suggests that GenAI can scaffold students towards deeper evaluative thinking, encouraging focus on substantive issues rather than surface errors. However, Noroozi et al. caution that reliance on 'polished, readily available' AI output may hinder students' own problem-solving and critical evaluation skills.

Recent literature further highlights how GenAI can promote critical thinking in feedback when well-integrated. The systematic review by Sichterman et al. (2025) finds that AI-driven interventions in peer learning can 'transform peer learning through responsive and adaptive interventions' while also warning of challenges like student over-reliance and the need for intentional instructional design. For example, Sichterman et al. summarise evidence from an EFL study showing that a ChatGPT-supported collaborative argumentation approach significantly improved students' critical thinking awareness and the quality of their arguments compared to a conventional method. This suggests GenAI's potential to scaffold higher-order thinking in argumentative contexts.

The study's results can be better understood when viewed through the lens of Social Cognitive Theory and Constructivist Learning Theory. From the perspective of SCT, the strong predictive role of AI self-efficacy in shaping students' perceptions confirms Bandura's (1986) argument that beliefs about one's own capabilities influence both motivation and cognitive engagement. Students who felt more confident in using AI tools not only reported more favourable views of AI's contribution to critical thinking and creativity but also seemed more willing to see these technologies as resources rather than threats. This supports the notion of reciprocal determinism, where confidence shapes behaviour, and positive experiences with AI may further reinforce confidence. Constructivist Learning Theory offers an additional lens for interpreting the disciplinary differences observed in this study. The more positive perceptions of STEM students towards AI's support for critical thinking reflect the alignment between computational tools and problem-solving approaches commonly emphasised in STEM curricula. By contrast, the more cautious stance of non-STEM students illustrates how disciplinary epistemologies, which often emphasise interpretive reasoning and originality, shape the way learners make meaning from AI-supported tasks (Jonassen, 1999). Taken together, these frameworks not only explain why self-efficacy and disciplinary orientation matter but also underscore the importance of designing learning environments that account for both individual beliefs and epistemological contexts.

Practical implications

The findings of this study carry direct implications for higher education institutions seeking to integrate GenAI into teaching and learning in ways that strengthen critical and creative thinking. A key starting point is the development of structured AI literacy curricula for undergraduate students. Such curricula should extend beyond technical training and include modules on critical evaluation of AI outputs, ethical use, and bias awareness. Embedding these elements in first-year courses would ensure that students develop both the confidence and reflective skills needed to engage productively with AI. Institutions could also design discipline-specific adaptations of AI literacy, where STEM students practice integrating AI into problem-solving and simulations. In contrast, non-STEM students focus on interrogating AI outputs for originality, perspective, and cultural sensitivity (Walter, 2024).

Equally important is faculty training. Instructors play a central role in shaping how students perceive and use AI, yet many faculty members report uncertainty about how to effectively integrate these tools into teaching (Holmes et al., 2022). Training programmes should, therefore, be twofold: (1) technical workshops that familiarise educators with current AI applications and their pedagogical affordances, and (2) pedagogical seminars that model how to use AI to scaffold learning rather than replacing cognitive effort. For example, faculty could be trained to design assignments that require students to critically annotate AI-generated drafts, compare machine responses with peer or expert feedback, or use AI as a brainstorming tool followed by iterative human refinement.

Finally, assessment strategies must be reconsidered in light of AI integration. Traditional written assignments risk becoming less effective if students over-rely on AI-generated content. Instead, educators should adopt assessment designs that foreground process and reflection. For instance, portfolio-based assessments could require students to document how they utilised AI during the different stages of a project, including justification of choices and critical commentary on AI-generated suggestions. Oral defences, peer-review exercises, and iterative submissions that capture the evolution of ideas can further help educators evaluate the depth of critical and creative engagement rather than the final product alone (Noroozi, Haddadian, et al., 2025).

Limitations and future directions

The use of perception-based survey data may be seen as a limitation, since reported attitudes do not always align with actual behaviours or cognitive performance (Noroozi, Alqassab, et al., 2025). However, perception data is particularly relevant for the research questions posed here. Students' beliefs about the role of GenAI are not merely subjective impressions but are key drivers of how they choose to adopt, engage with, and benefit from these tools in educational settings. Also, the cultural and geographical context of this study must be considered when interpreting the findings. The data were collected from Saudi undergraduates, a group embedded in specific cultural, educational, and policy environments that shape both the integration of AI tools and students' perceptions of them. The findings may not be directly generalisable to other regions with different cultural values and levels of AI integration. Future research should include cross-cultural or multinational samples to

compare how students from diverse educational ecosystems perceive AI's cognitive and creative affordances. Finally, there is the possibility of response bias in measuring AI-related self-efficacy. Students may have either overestimated or underestimated their competence due to social desirability, uncertainty about what constitutes 'effective use', or lack of prior experience with advanced AI tools. These biases could influence the observed relationships between self-efficacy and perceived benefits of AI. Triangulating self-reports with behavioural data, such as digital traces of AI usage or performance-based tasks, would provide a more robust measure of self-efficacy in future studies. Despite these limitations, the study makes an important contribution by highlighting the interplay of self-efficacy and disciplinary differences in shaping how students perceive AI's role in higher-order cognition.

Conclusion

While students generally viewed GenAI as moderately supportive of higher-order thinking and creative processes, their perceptions were significantly shaped by individual and contextual factors. STEM students were more inclined than their non-STEM counterparts to view GenAI as beneficial for enhancing critical thinking. This disciplinary difference underscores how the epistemological orientation and pedagogical practices of specific fields shape the ways students engage with AI technologies. Interestingly, perceptions of AI's creative affordances were consistent across both STEM and non-STEM disciplines. The present study calls for a balanced and pedagogically grounded integration of AI in higher education. Taken together, educators and institutions must prioritise the development of students' AI literacy and self-regulatory capacities. Such efforts will not only maximise the potential of AI as a cognitive and creative scaffold but also ensure that its use aligns with the broader goals of intellectual autonomy, critical inquiry, and ethical scholarship.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributor

Atiyah A. Alghamdi is an Assistant Professor of Educational Leadership and Policy Studies at King Saud University with over 20 years of experience in public and higher education. He has served in multiple leadership and advisory roles across national ministries and universities, led major development initiatives, and is an ICF-accredited leadership coach. He currently serves as Vice Dean for Educational and Academic Affairs.

ORCID

Atiyah A. Alghamdi http://orcid.org/0000-0002-2002-673X

Consent to participate

All participants provided written informed consent prior to participating.

Data availability statement

The data is available from the corresponding author upon reasonable request.

Ethics approval

The ethical approval/IRB permission was granted by the King Saud University, Saudi Arabia. (Application No. KSU-HE-25–473).

References

Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Prentice-Hall.

Banihashem, S. K., Kerman, N. T., Noroozi, O., Moon, J., & Drachsler, H. (2024). Feedback sources in essay writing: Peer-generated or AI-generated feedback? *International Journal of Educational Technology in Higher Education*, *21*(23). https://doi.org/10.1186/s41239-024-00455-4

Banihashem, S. K., Noroozi, O., Khosravi, H., Schunn, C. D., & Drachsler, H. (2025). Pedagogical framework for hybrid intelligent feedback. *Innovations in Education and Teaching International*, 1–17. https://doi.org/10.1080/14703297.2025.2499174

Becher, T., & Trowler, P. (2001). *Academic tribes and territories*. McGraw-Hill Education (UK).

Chan, C. K. Y., & Hu, W. (2023). Students' voices on generative AI: Perceptions, benefits, and challenges in higher education. *International Journal of Educational Technology in Higher Education*, *20*(1), 43. https://doi.org/10.1186/s41239-023-00411-8

Cohen, J. (2013). *Statistical power analysis for the behavioral sciences* (2nd ed.). Routledge.

Facione, P. A., Facione, N. C., & Sanchez, C. A. (1994). Critical thinking disposition as a measure of competent clinical judgment: The development of the California critical thinking disposition inventory. *Journal of Nursing Education*, *33*(8), 345–350. https://doi.org/10.3928/0148-4834-19941001-05

Falebita, O. S., & Kok, P. J. (2024). Artificial intelligence tools usage: A structural equation modeling of undergraduates' technological readiness, self-efficacy and attitudes. *Journal for STEM Education Research*, *8*(2), 257–282. https://doi.org/10.1007/s41979-024-00132-1

Field, A. (2018). *Discovering statistics using IBM SPSS statistics* (5th ed.). Sage.

Gerlich, M. (2025). AI tools in society: Impacts on cognitive offloading and the future of critical thinking. *Societies*, *15*(1), 6. https://doi.org/10.3390/soc15010006

Gonsalves, C. (2024). Generative AI's impact on critical thinking: Revisiting Bloom's taxonomy. *Journal of Marketing Education*. https://doi.org/10.1177/02734753241305980

Granda, B. S., Inzhivotkina, Y., Apolo, M. F. I., & Fajardo, J. G. U. (2024). Educational innovation: Exploring the potential of generative artificial intelligence in cognitive schema building. *Edutec, Revista Electrónica de Tecnología Educativa*, *89*(89), 44–63. https://doi.org/10.21556/edutec.2024.89.3251

Guilford, J. P. (1967). *The nature of human intelligence*. McGraw-Hill.

Holmes, W., Bialik, M., & Fadel, C. (2022). *Artificial intelligence in education: Promises and implications for teaching and learning*. Center for Curriculum Redesign.

Jonassen, D. H. (1999). Designing constructivist learning environments. In C. M. Reigeluth (Ed.), *Instructional-design theories and models* (pp. 215–239). Lawrence Erlbaum.

Karwowski, M., Jankowska, D. M., Brzeski, A., Czerwonka, M., Gajda, A., Lebuda, I., & Beghetto, R. A. (2020). Delving into creativity and learning. In V. P. Glăveanu, I. J. Ness, & C. de Saint Laurent (Eds.), *Creative learning in digital and virtual environments* (pp. 7–29). Routledge.

Kirkpatrick, K. (2023). Can AI demonstrate creativity? *Communications of the ACM*, *66*(2), 21–23. https://doi.org/10.1145/3575665

Kuhn, D. (1991). *The skills of argument*. Cambridge University Press.

Lockhart, E. N. (2025). Creativity in the age of AI: The human condition and the limits of machine generation. *Journal of Cultural Cognitive Science*, *9*(1), 83–88. https://doi.org/10.1007/s41809-024-00158-2

Marengo, A., Pagano, A., Pange, J., & Soomro, K. A. (2024). The educational value of artificial intelligence in higher education: A 10-year systematic literature review. *Interactive Technology and Smart Education*, *21*(4), 625–644. https://doi.org/10.1108/ITSE-11-2023-0218

Noroozi, O., Alqassab, M., Taghizadeh Kerman, N., Banihashem, S. K., & Panadero, E. (2025). Does perception mean learning? Insights from an online peer feedback setting. *Assessment & Evaluation in Higher Education*, *50*(1), 83–97. https://doi.org/10.1080/02602938.2024.2345669

Noroozi, O., Haddadian, G., Gao, X., Schunn, C., Alqassab, M., & Banihashem, S. K. (2025). The value of GenAI for peer feedback provision: Student perceptions and impacts. *International Journal of Educational Technology in Higher Education*, *22*(61). https://doi.org/10.1186/s41239-025-00558-6

Premkumar, P., Yatigammana, M. R. K. N., & Kannangara, S. (2024). Impact of generative AI on critical thinking skills in undergraduates: A systematic review. *The Journal of Desk Research Review and Analysis*, *2*(1), 199–215. https://doi.org/10.4038/jdrra.v2i1.55

Runco, M. A. (2012). *Creativity: Theories and themes: Research, development, and practice*. Elsevier Academic Press.

Shanto, S. S., Ahmed, Z., & Jony, A. I. (2024). Enriching learning process with generative AI: A proposed framework to cultivate critical thinking in higher education using chat GPT. *Tuijin Jishu/Journal of Propulsion Technology*, *45*(1), 3019–3029.

Sichterman, B., Noroozi, O., Boetje, J., van Ginkel, S., Khosravi, H., & Versendaal, J. (2025). Supporting peer learning with artificial intelligence: A systematic literature review. *Innovations in Education and Teaching International*, *62*(5), 1648–1664. https://doi.org/10.1080/14703297.2025.2530118

Soomro, K. A., Ansari, M., Bughio, I., & Nasrullah, N. (2024). Examining gender and urban-rural divide in digital competence among university students. *International Journal of Learning Technology*, *19*(3), 380–393. https://doi.org/10.1504/IJLT.2024.142512

Soomro, K. A., Yousuf Zai, S., & Jafri, I. H. (2015). Competence and usage of web 2.0 technologies by higher education faculty. *Educational Media International*, *52*(4), 284–295. https://doi.org/10.1080/09523987.2015.1095522

Walter, Y. (2024). Embracing the future of artificial intelligence in the classroom: The relevance of AI literacy, prompt engineering, and critical thinking in modern education. *International Journal of Educational Technology in Higher Education*, *21*(1), 15. https://doi.org/10.1186/s41239-024-00448-3

Wang, F., Li, N., Cheung, A. C., & Wong, G. K. (2025). In GenAI we trust: An investigation of university students' reliance on and resistance to generative AI in language learning. *International Journal of Educational Technology in Higher Education*, *22*(1), 59. https://doi.org/10.1186/s41239-025-00547-9

Xu, W., & Ouyang, F. (2022). The application of AI technologies in STEM education: A systematic review from 2011 to 2021. *International Journal of STEM Education*, *9*(1), 59. https://doi.org/10.1186/s40594-022-00377-5

Yusuf, A., Pervin, N., & Román-González, M. (2024). Generative AI and the future of higher education: A threat to academic integrity or reformation? Evidence from multicultural perspectives. *International Journal of Educational Technology in Higher Education*, *21*(1). https://doi.org/10.1186/s41239-024-00453-6